

Consumption and Saving

Casper Moes, hwd362

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1 MPC with infinite horizon

a Deriving the Euler equation

a.1 Bellman equation

The (representative) consumer faces the following utility maximization problem

$$\begin{aligned} \max_{c_t, a_{t+1}} U_t &\equiv \sum_{k=0}^{\infty} \beta^{t+k} \frac{c_{t+k}^{1-\sigma}}{1-\sigma} \\ \text{s.t.} \\ c_t + a_{t+1} &= y_t + (1+r)a_t \\ \lim_{T \rightarrow \infty} \left(\prod_{t=0}^T (1+r)^{-1} \right) a_{T+1} &\geq 0, \end{aligned}$$

where a_0 is given and $\sigma > 1$. First I derive the Bellman equation explicitly (for an arbitrary starting index t , which will usually be set to $t = 0$)

$$\begin{aligned} V(a_t) &\equiv \max_{\{c_{t+k}, a_{t+1+k}\}_{k=0}^{\infty}} \underbrace{\left\{ \sum_{k=0}^{\infty} \beta^k \frac{c_{t+k}^{1-\sigma}}{1-\sigma} \right\}}_{U_t} \\ &= \max_{\{c_{t+k}, a_{t+1+k}\}_{k=0}^{\infty}} \left\{ \frac{c_t^{1-\sigma}}{1-\sigma} + \beta \sum_{k=0}^{\infty} \beta^k \frac{c_{t+1+k}^{1-\sigma}}{1-\sigma} \right\} \\ &= \max_{c_t, a_{t+1}} \left\{ \frac{c_t^{1-\sigma}}{1-\sigma} + \beta V(a_{t+1}) \right\} \\ &= \max_{a_{t+1}} \left\{ \frac{(y_t + (1+r)a_t - a_{t+1})^{1-\sigma}}{1-\sigma} + \beta V(a_{t+1}) \right\} \end{aligned}$$

The first order condition for an internal optima (for the right hand side) is given by

$$\underbrace{(y_t + (1+r)a_t - a_{t+1})^{-\sigma}}_{c_t} = \beta V_a(a_{t+1}), \quad (1)$$

where we have defined $V_a(a_t) \equiv \frac{\partial V}{\partial a_t}$. By the envelope theorem (assuming V is differentiable) we have

$$V_a(a_{t+1}) = (1+r)c_{t+1}^{-\sigma}. \quad (2)$$

a.2 Proof of envelope theorem

Define the optimal asset policy $a_{t+1}^* = g(a_t)$ as the function that solves the Bellman equation:

$$V(a_t) = u(y_t + (1+r)a_t - g(a_t)) + \beta V(g(a_t))$$

differentiating wrt. a_t yields

$$\begin{aligned}
 V_a(a_t) &= \underbrace{u'(c_t)((1+r) - g'(a_t))}_{\text{gain due to increased current consumption}} + \underbrace{\beta V_a(g(a_t))g'(a_t)}_{\text{gain due to increased savings}} \\
 &= (1+r)u'(c_t) + g'(a_t) \underbrace{(\beta V_a(g(a_t)) - u'(c_t))}_{=0 \text{ (because of FOC)}}.
 \end{aligned}$$

Q.E.D.

Intuitively $(\beta V_a(g(a_t)) - u'(c_t)) = 0$ because the marginal gain of spending is equal the marginal gain of saving at the maximum. therefore shifting this a little bit does not matter for the value function. So basically the increase in assets is distributed as utility gain due to increased current consumption and increased future savings, but disregarding the distribution over current and future consumption as this gives the same marginal utility at the maximum, which is $(1+r)u'(c_t)$.

a.3 The Euler equation

Combining (1) and (2) we obtain the well known Euler equation

$$c_t^{-\sigma} = \beta(1+r)c_{t+1}^{-\sigma}, \quad (3)$$

which is a necessary condition for an internal optimal consumption path.

b The budget constraint

b.1 Transversality condition

From the maximization problem we have the no ponzi-scheme condition

$$\lim_{T \rightarrow \infty} \left(\prod_{t=0}^T (1+r)^{-1} \right) a_{T+1} = \lim_{T \rightarrow \infty} ((1+r)^{-(T+1)}) a_{T+1} \geq 0,$$

Because the utility function is monotone in consumption

$$\frac{\partial U_t}{\partial c_{t+k}} > 0, \quad \forall k \geq 0,$$

and the consumer gains no direct utility from assets

$$\frac{\partial U_t}{\partial a_{t+k}} = 0, \quad \forall k \geq 0,$$

we conclude that the consumer would not increase savings indefinitely, and such the transversality condition is born

$$\lim_{T \rightarrow \infty} ((1+r)^{-(T+1)}) a_{T+1} = 0. \quad (4)$$

b.2 Budget constraint

Iterating forward the budget constraint yields

$$\begin{aligned}
 a_t &= (1+r)^{-1} (c_t - y_t) + (1+r)^{-1} a_{t+1} \\
 &= (1+r)^{-1} (c_t - y_t) + (1+r)^{-1} \left((1+r)^{-1} (c_{t+1} - y_{t+1}) + (1+r)^{-1} a_{t+2} \right) \\
 &= (1+r)^{-1} (c_t - y_t) + (1+r)^{-2} (c_{t+1} - y_{t+1}) + (1+r)^{-2} a_{t+2} \\
 &= (1+r)^{-1} (c_t - y_t) + (1+r)^{-1} + (1+r)^{-2} (c_{t+1} - y_{t+1}) + (1+r)^{-3} (c_{t+2} - y_{t+2}) + (1+r)^{-3} a_{t+3} \\
 &= \sum_{k=0}^T (1+r)^{-(k+1)} (c_{t+k} - y_{t+k}) + (1+r)^{-(T+1)} a_{T+1}.
 \end{aligned}$$

Taking the limit $T \rightarrow \infty$, evaluating at the starting period $t = 0$ and using (4) yields

$$\begin{aligned}
 a_0 &= \sum_{k=0}^{\infty} (1+r)^{-(k+1)} (c_k - y_k) + \lim_{T \rightarrow \infty} (1+r)^{-(T+1)} a_{T+1} \\
 &= \sum_{k=0}^{\infty} (1+r)^{-(k+1)} (c_k - y_k) \iff \\
 &\quad \sum_{k=0}^{\infty} (1+r)^{-k} c_k = (1+r)a_0 + h_0,
 \end{aligned} \tag{5}$$

where $h_0 \equiv \sum_{k=0}^{\infty} (1+r)^{-k} y_k$ is the present value of human capital / income stream. That is the present value of consumption is equal to the value of period-0 assets and the present value of income.

c Consumption function

Firstly we rewrite the Euler equation (3)

$$c_t^{-\sigma} = \beta(1+r)c_{t+1}^{-\sigma} \iff c_t = (\beta(1+r))^{-1/\sigma} c_{t+1},$$

and derive the consumption at every period in terms of initial consumption c_0 :

$$c_t = (\beta(1+r))^{1/\sigma} c_{t-1} = (\beta(1+r))^{t/\sigma} c_0. \tag{6}$$

Substituting (6) for all k into the budget constraint (5) yields the consumption function

$$\begin{aligned}
 \left[\sum_{k=0}^{\infty} \left(\frac{(\beta(1+r))^{1/\sigma}}{1+r} \right)^k \right] c_0 &= \left(1 - \frac{(\beta(1+r))^{1/\sigma}}{1+r} \right)^{-1} c_0 = (1+r)a_0 + h_0 \iff \\
 c_0 &= \left(1 - \frac{(\beta(1+r))^{1/\sigma}}{1+r} \right) ((1+r)a_0 + h_0).
 \end{aligned}$$

d Marginal Propensities to Consume (MPCs)

2 The Ramsey model

a Firms

The firm maximises profit in each period (non-dynamically) subject to a Cobb-Douglas production technology. Formally it solves:

$$\max_{K_t, L_t} \left\{ \pi(K_t, L_t) = AK_t^\alpha L_t^{1-\alpha} - w_t L_t - r_t K_t \right\}, \quad (7)$$

$\alpha \in (0, 1)$. The problem (7) has the following necessary first order conditions for an internal optimum

$$\frac{\partial \pi}{\partial K_t} = A\alpha K_t^{\alpha-1} L_t^{1-\alpha} - r_t = 0 \iff \alpha A k_t^{\alpha-1} = r_t, \quad (8)$$

$$\frac{\partial \pi}{\partial L_t} = AK_t^\alpha (1-\alpha) L_t^{-\alpha} - w_t = 0 \iff (1-\alpha) A k_t^\alpha = w_t. \quad (9)$$

The labor-income share is then

$$\frac{w_t L_t}{AK_t^\alpha L_t^{1-\alpha}} = \frac{w_t}{A k_t^\alpha} = \frac{(1-\alpha) A k_t^\alpha}{A k_t^\alpha} = (1-\alpha).$$

b Empirical labor-income

c Households

Households solve the following utility maximisation problem

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} U = \sum_{t=0}^{\infty} \beta^t (1+n)^t \frac{c_t^{1-\sigma}}{1-\sigma}$$

s.t.

$$(1+n)a_{t+1} = a_t R_t + w_t - c_t$$

where $\sigma > 0$, $\sigma \neq 1$, $R_t \equiv 1 + r_t - \delta$ and $a_0 = k_0 > 0$ given. The problem has the following Lagrangian

$$\mathcal{L} = \left[\sum_{t=0}^{\infty} \beta^t (1+n)^t \frac{c_t^{1-\sigma}}{1-\sigma} + \lambda_t (a_t R_t + w_t - c_t - (1+n)a_{t+1}) \right],$$

with the first order conditions

$$\frac{\partial \mathcal{L}}{\partial c_t} = \beta^t (1+n)^t c_t^{-\sigma} - \lambda_t = 0 \iff \beta^t (1+n)^t c_t^{-\sigma} = \lambda_t,$$

$$\frac{\partial \mathcal{L}}{\partial a_{t+1}} = -(1+n)\lambda_t + \lambda_{t+1} R_{t+1} = 0 \iff \frac{1}{1+n} \lambda_{t+1} R_{t+1} = \lambda_t$$

substituting the multipliers for $\lambda_t = \beta^t(1+n)^t c_t^{-\sigma}$ and $\lambda_{t+1} = \beta^{t+1}(1+n)^{t+1} c_{t+1}^{-\sigma}$ yields the consumption Euler equation

$$\begin{aligned} \frac{1}{1+n} \left(\beta^{t+1}(1+n)^{t+1} c_{t+1}^{-\sigma} \right) R_{t+1} &= \beta^t(1+n)^t c_t^{-\sigma} \\ &\iff \\ c_t^{-\sigma} &= \beta(1+r) c_{t+1}^{-\sigma}. \end{aligned} \tag{10}$$

d Equilibrium

We impose market clearing conditions

$$\begin{aligned} k_t &\equiv \frac{K_t}{L_t} = a_t, \\ c_t = y_t &\equiv \frac{AK_t^\alpha L_t^{1-\alpha}}{L_t} = Ak_t^\alpha, \end{aligned}$$

for all $t \in \{0, 1, \dots\}$.

e Does the model hold up?